Course Name: Physics Lab
Course Code: PH-102

Contact Hours/Week: 02P Course Credits: **01**

Course Objectives:

- To gain practical knowledge by applying the experimental methods to correlate with the theory.
- To learn the usage of electrical and optical systems for various measurements.
- Apply the analytical techniques and graphical analysis to the experimental data.
- To develop intellectual communication skills and discuss the basic principles of scientific concepts in a group.

List of Experiments

1. To determine the specific resistance of a material wire using a post office box.
2. To find the area of a rectangle (or height of an inaccessible object) using a sextant.
3. Conversion of a galvanometer into Ammeter and Voltmeter of given range.
4. To verify the inverse square law of magnetism.
5. Study the variation of magnetic field with distance along the axis of a circular coil carrying current and to find the radius of the coil.
6. To determine the refractive index of a glass/ liquid (water) using Spectrometer.
7. To determine the wavelength of light using Newton's ring apparatus.
8. To verify the inverse square law for the intensity of radiation from a source of light.
9. To determine the wavelength of the Laser light using diffraction method.
10. To find magnifying power of a telescope by linear method.
11. To measure Young's modulus by bending of beam method.
12. Study of the attenuation and propagation characteristics of an optical fiber cable.
13. Other experiments as and when made available time to time.

Course Outcomes:

Upon successful completion of the course, the students will be able to

CO1: Handle equipments and take measurements and record data techniques for the experiments.

CO2: Experimentally realize the physical phenomenon/ effects.

CO3: Use different systems and instruments to measuring parameters with precision.

CO4: Develop basic communication skills through working in groups in performing the laboratory experiments and by interpreting the results.